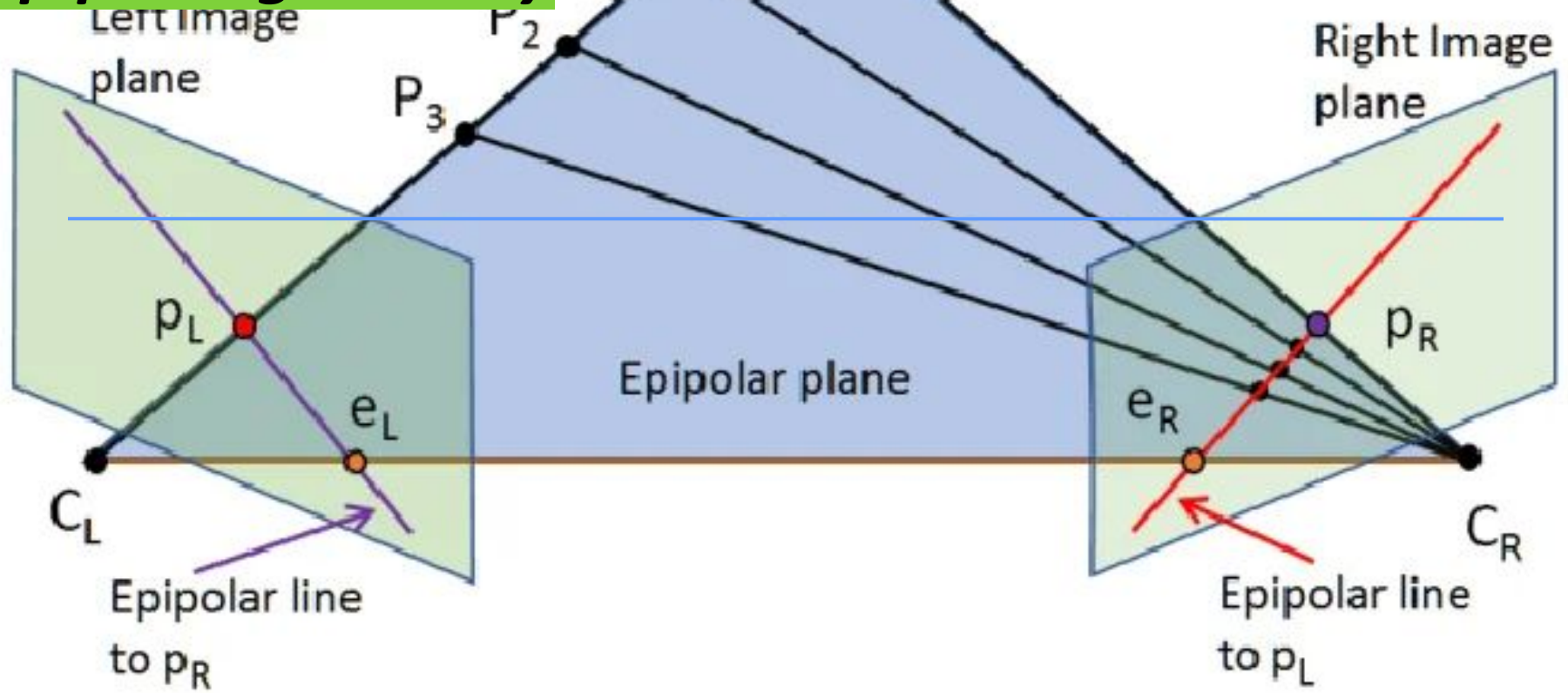


COMPUTER VISION

Epipolar geometry



Depth from disparity

image $I(x,y)$

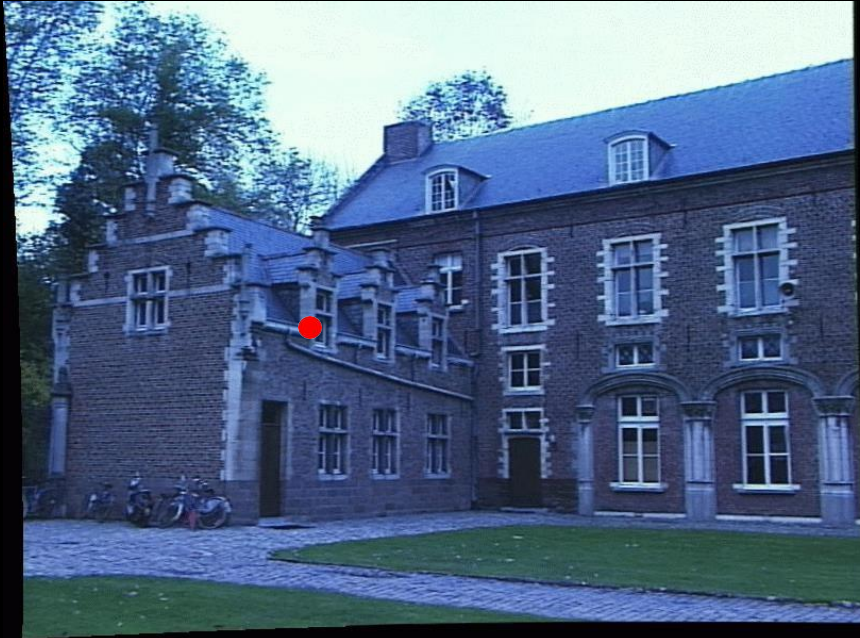


image $I'(x,y)$



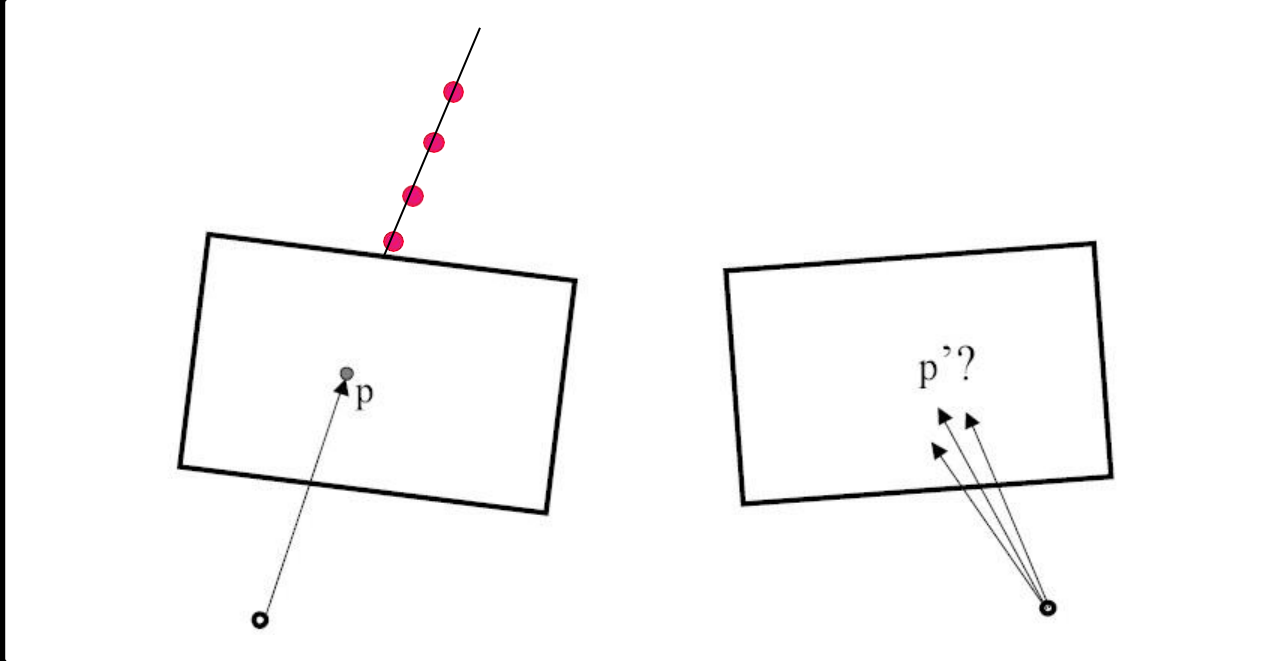
Depth from disparity

image $I(x,y)$

Disparity map $D(x,y)$



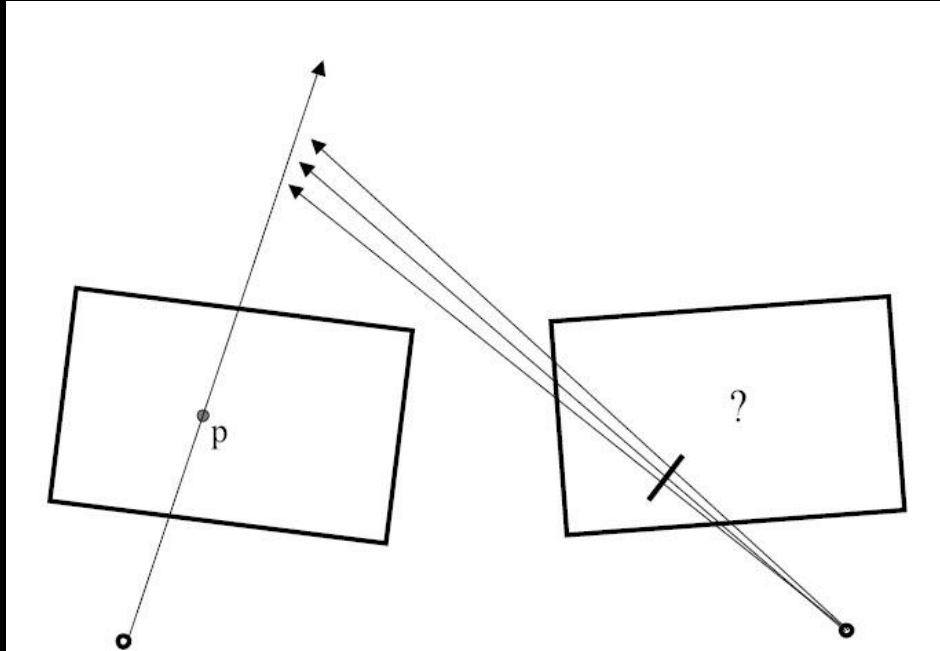
Stereo correspondence constraints



Given p in left image, where can corresponding point p' be?

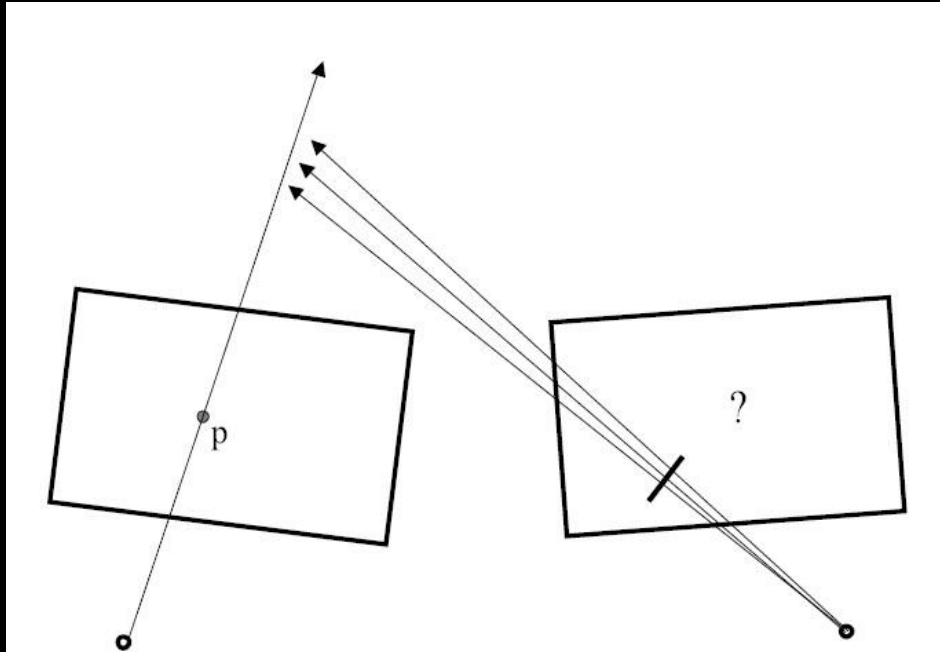
Stereo correspondence constraints

Remember:
in perspective
projection, lines
project into
lines.

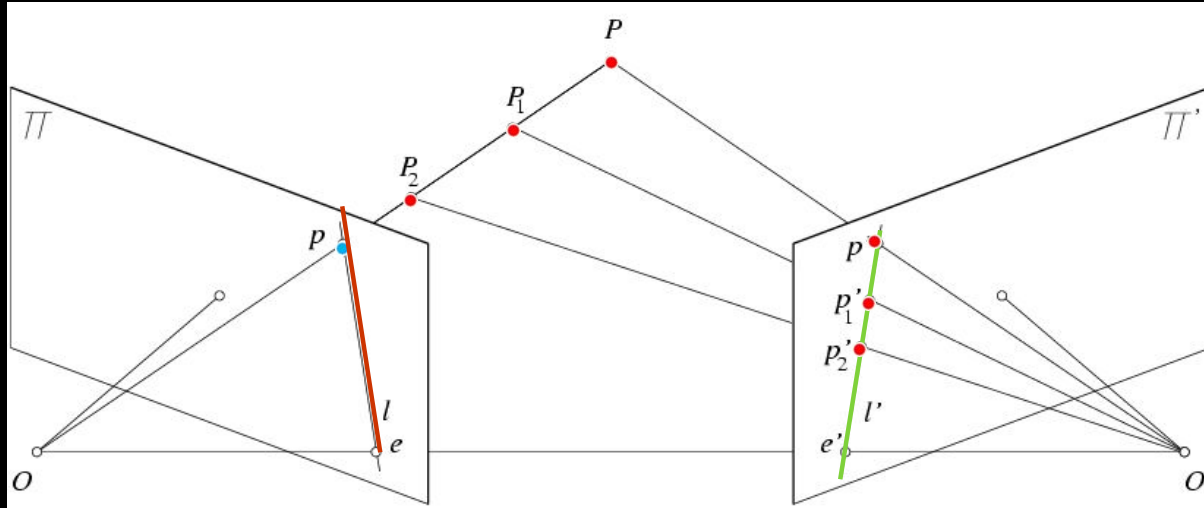


Stereo correspondence constraints

So the **line** containing the center of projection and the point P in the left image must project to a **line** in the right image.



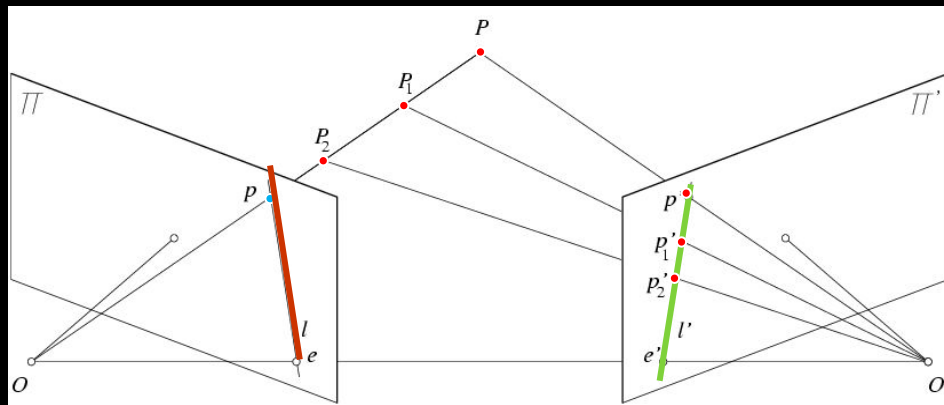
Epipolar constraint



Geometry of two views constraints where the corresponding pixel for some image point in the first view must occur in the second view.

Epipolar geometry: Terms

- **Baseline**: line joining the camera centers
- **Epipolar plane**: plane containing baseline and world point
- **Epipolar line**: intersection of epipolar plane with the image plane – come in pairs
- **Epipole**: point of intersection of baseline with image plane



Why is the epipolar constraint useful?

Epipolar constraint

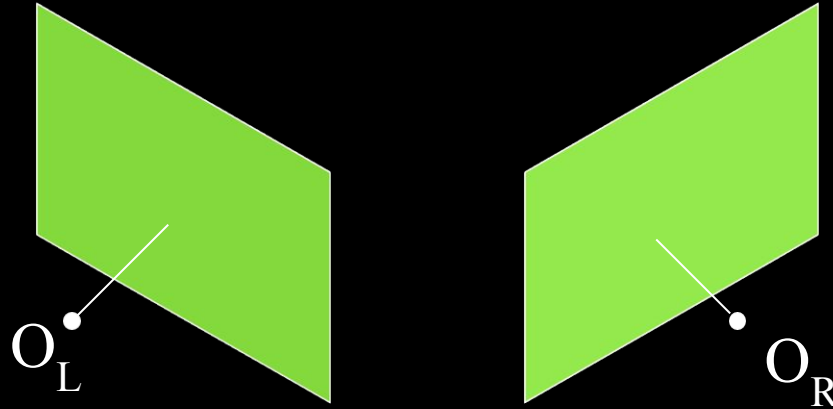


The *epipolar constraint* reduces the correspondence problem to a 1D search along an epipolar line.

Image from Andrew Zisserman

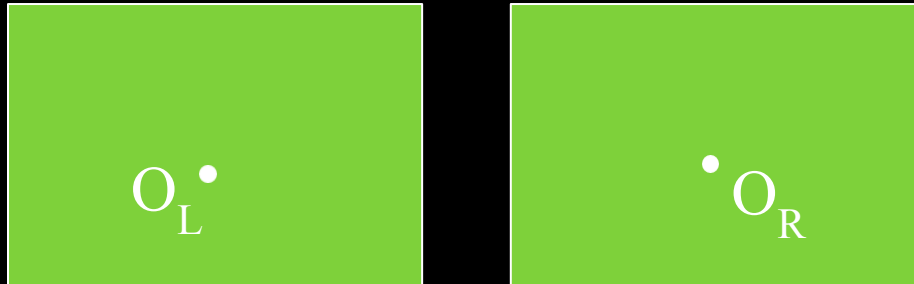
What do the epipolar lines look like?

1.



General case:
converged
camera

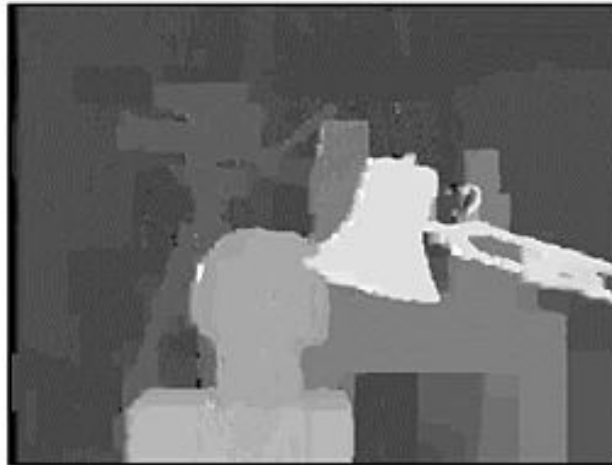
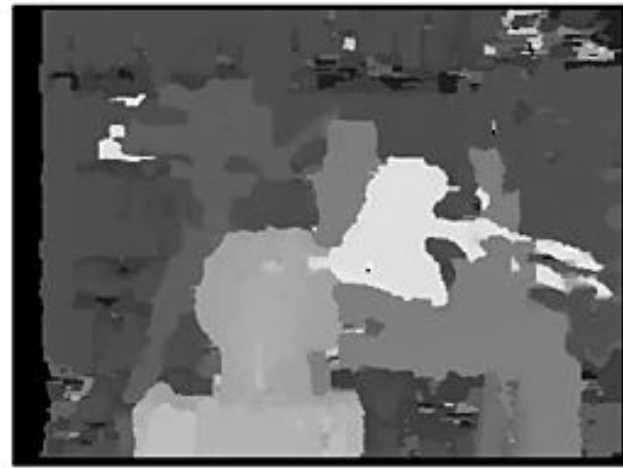
2.



Special case:
Parallel optical
axis

COMPUTER VISION

Stereo correspondence



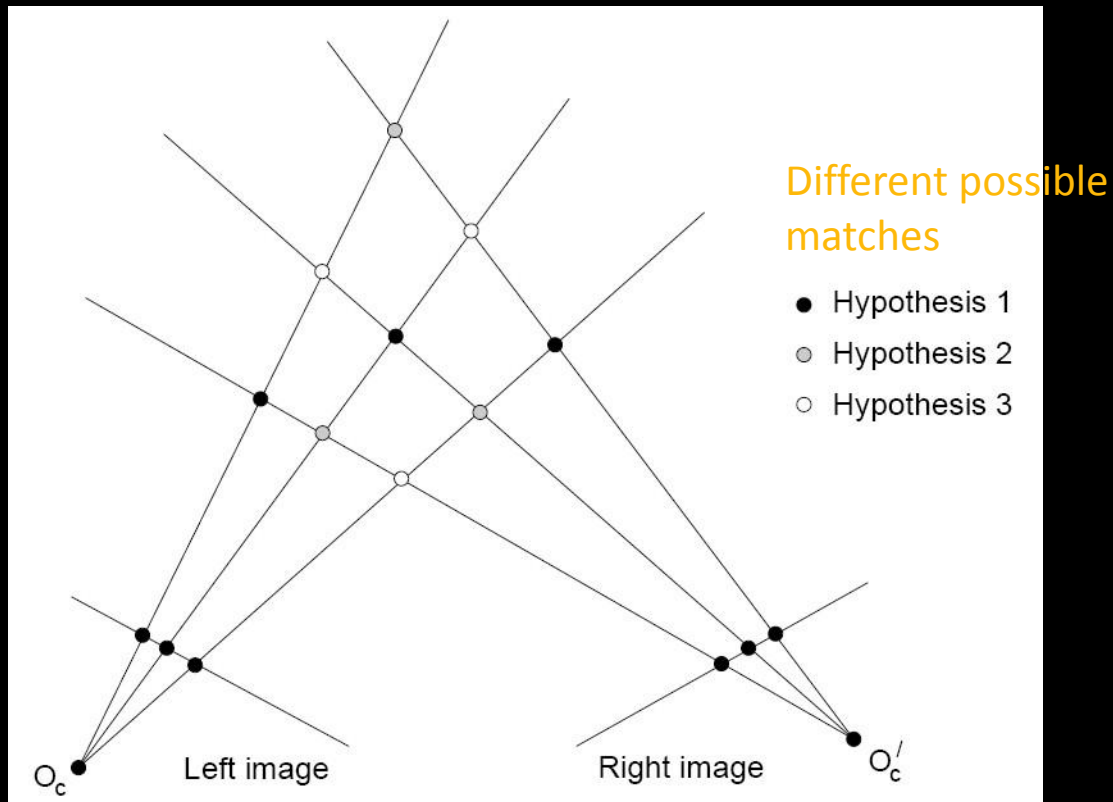
For now assume parallel image planes

- Assume parallel (coplanar) image planes...
- Assume same focal lengths...
- Assume epipolar lines are horizontal...
- Assume epipolar lines are at the same y location in the image (different x same y)...
- That's a lot of assuming, but it allows us to move to the correspondence problem – which you will be solving!

Correspondence problem

Multiple match hypotheses satisfy *epipolar constraint*, but which is correct?

Figure from Gee & Cipolla 1999



Correspondence problem

Beyond the hard constraint of epipolar geometry, there are “soft” constraints to help identify corresponding points

- Similarity - pixels in left should be same as pixels in right
- Uniqueness - not more than one matching
- Ordering - ordering should be same
- Disparity gradient is limited - depth does not change too quickly

Correspondence problem

Beyond the hard constraint of epipolar geometry, there are “soft” constraints to help identify corresponding points

- **Similarity**

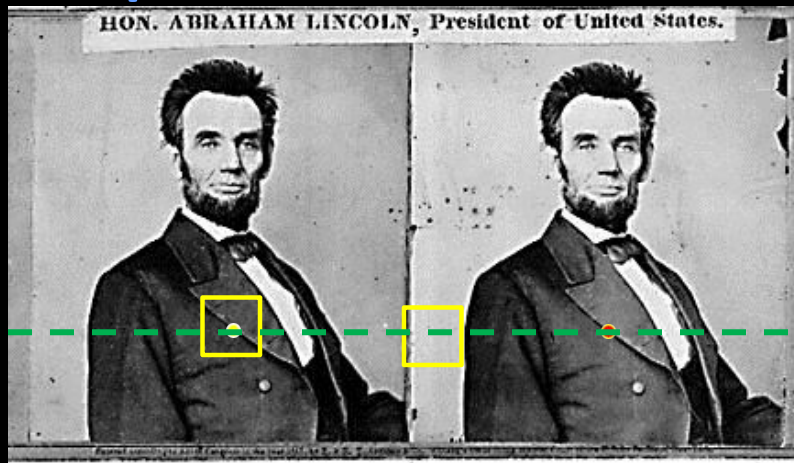
- Uniqueness
- Ordering
- Disparity gradient is limited

Correspondence problem

To find matches in the image pair, we will assume

- Most scene points visible from both views
- Image regions for the matches are similar in appearance

Dense correspondence search

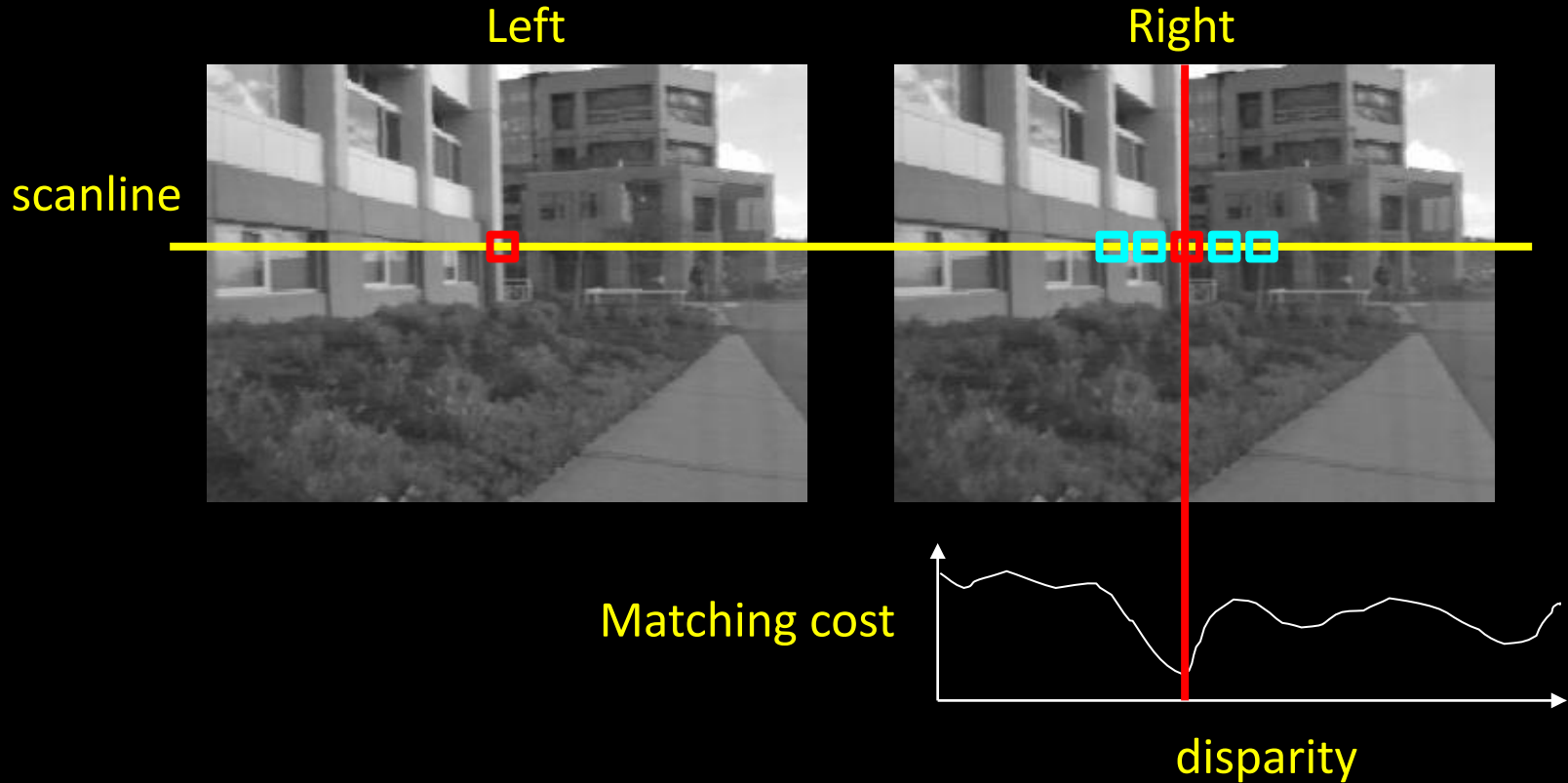


For each pixel / window in the left image

- Compare with every pixel / window on same epipolar line in right image
- Pick position with minimum match cost (e.g., SSD, normalized correlation)

Adapted from Li Zhang

Correspondence search: (dis)similarity constraint

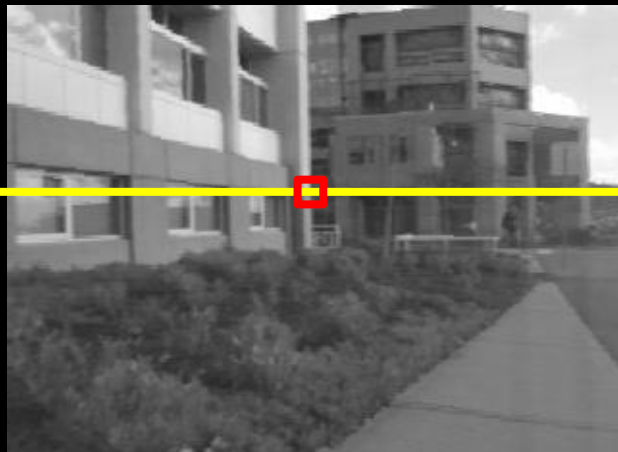


Correspondence search: (dis)similarity constraint

Left

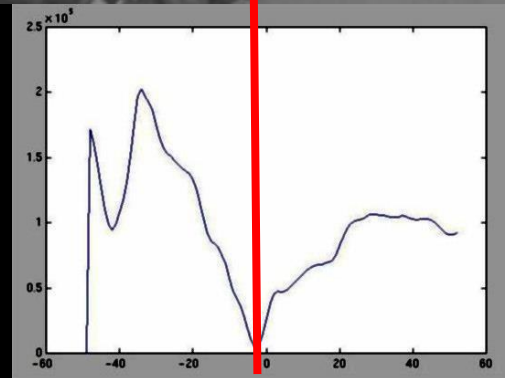
Right

scanline



Take 2 pixel windows
Overlap them
Subtract them
Square the differences
Sum them up

SSD

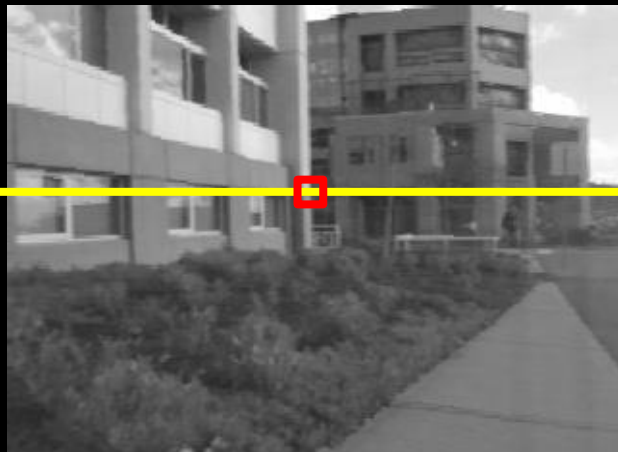


Correspondence search: (dis)similarity constraint

Left

Right

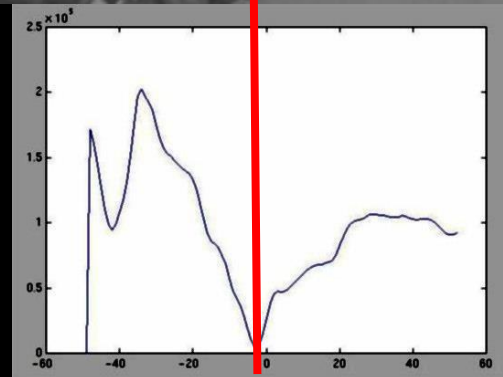
scanline



What if one image is brighter than the other?

Subtraction?? will lead to squared difference error

SSD

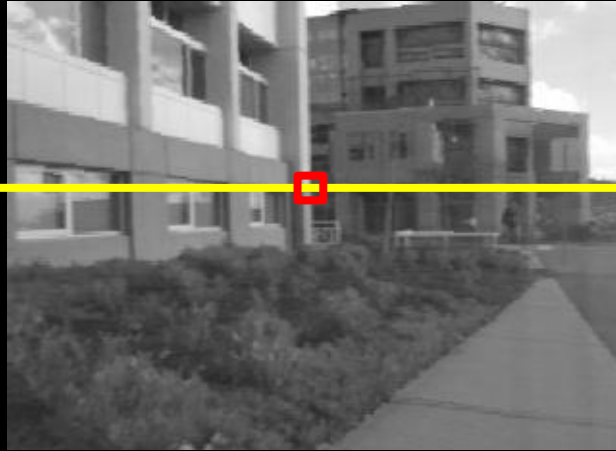


Correspondence search: similarity constraint

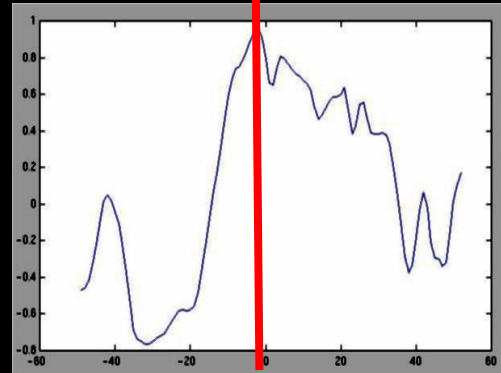
Left

Right

scanline



Normalized
correlation



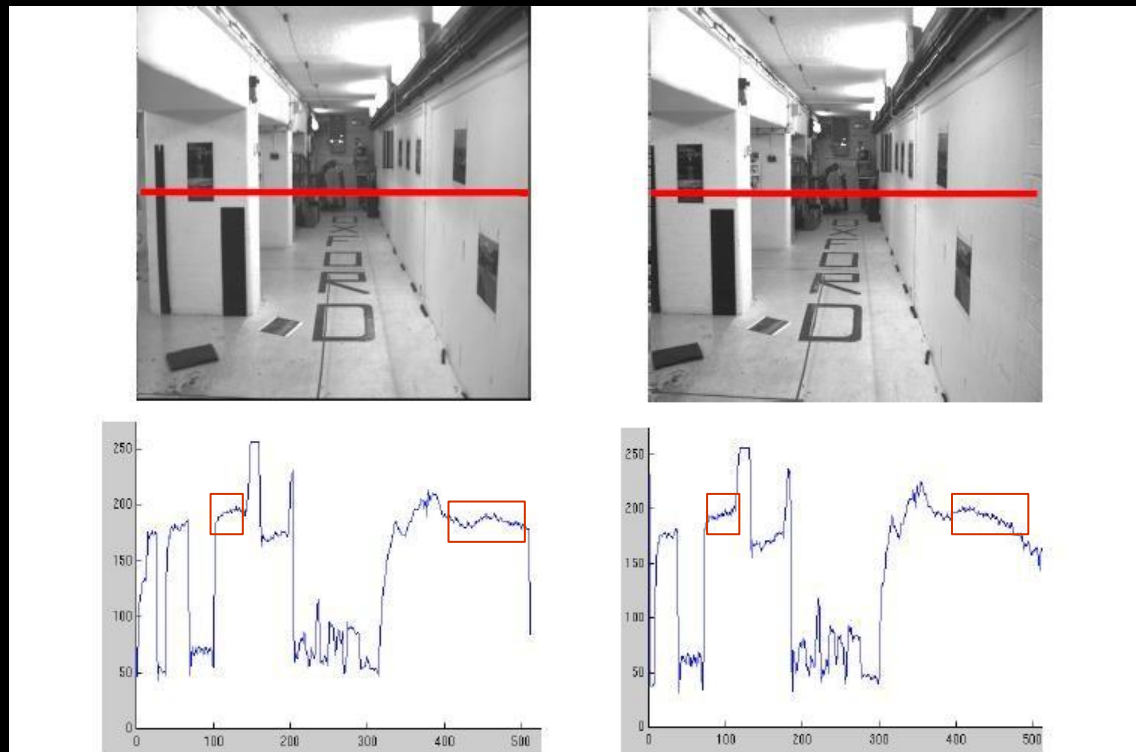
Correspondence problem



Source: Andrew Zisserman

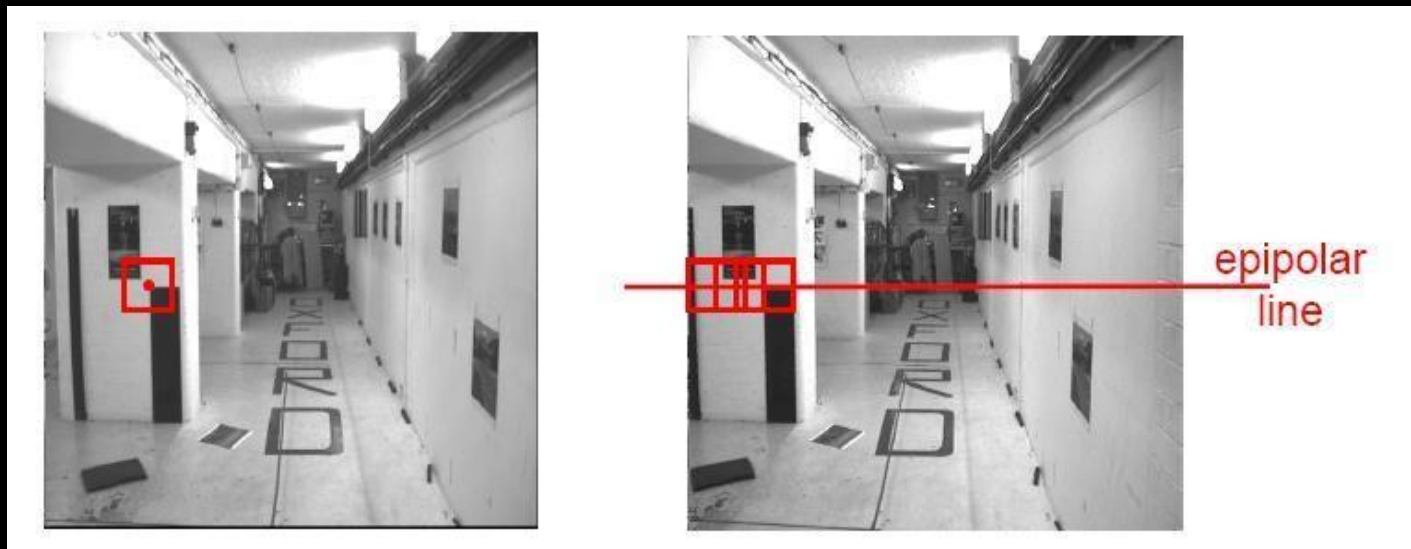
Correspondence problem

Intensity
profiles



Source: Andrew Zisserman

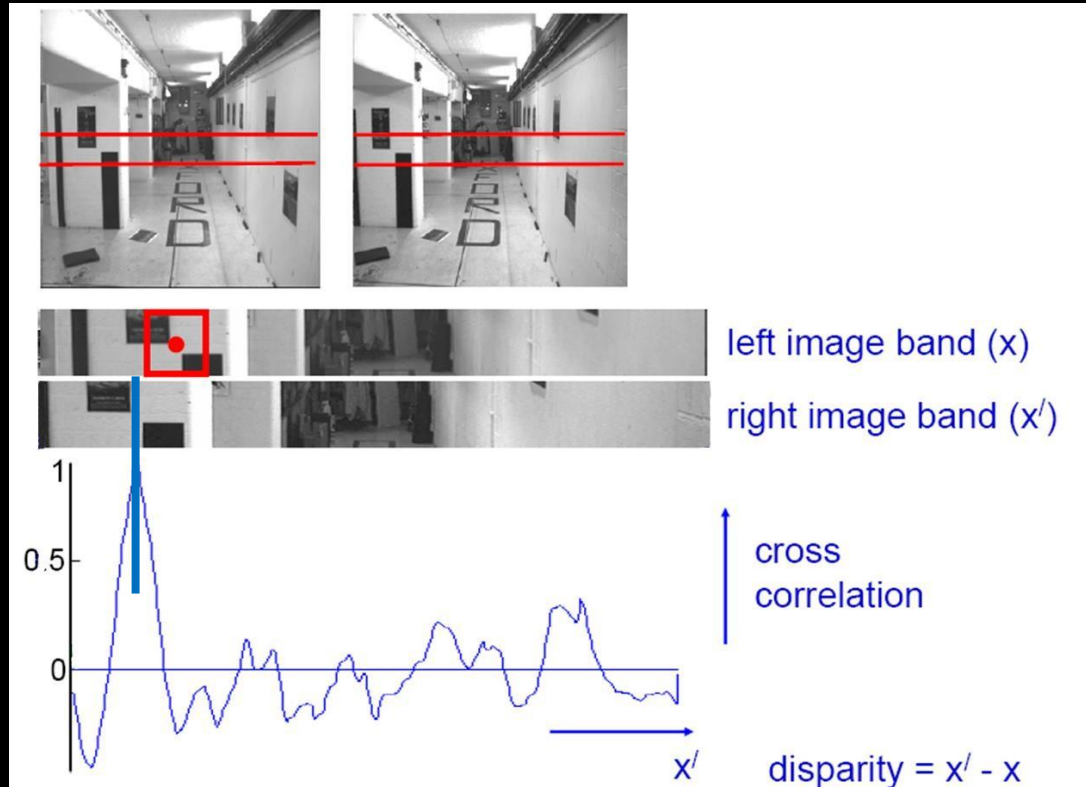
Correspondence problem



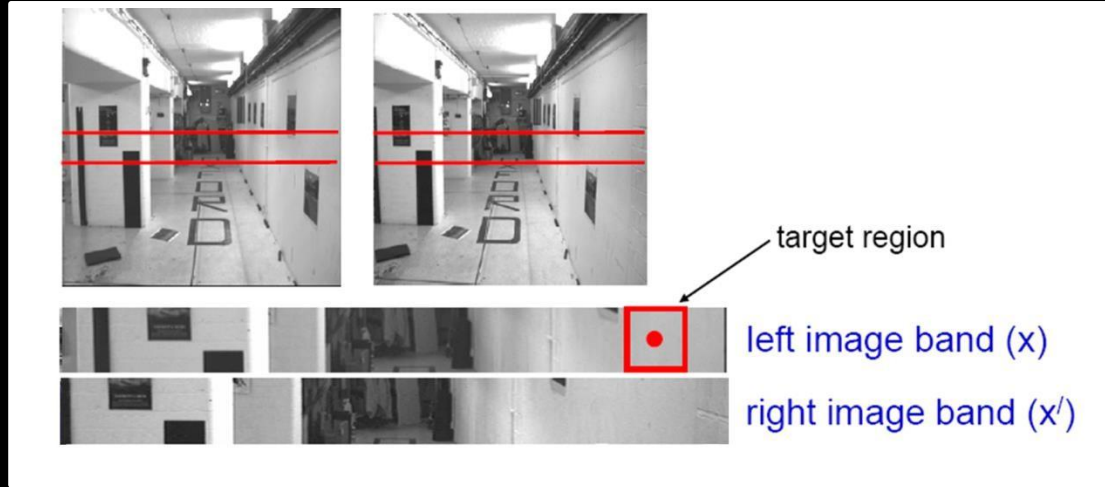
Neighborhoods of corresponding points
are similar in intensity patterns.

Source: Andrew Zisserman

Correlation-based window matching

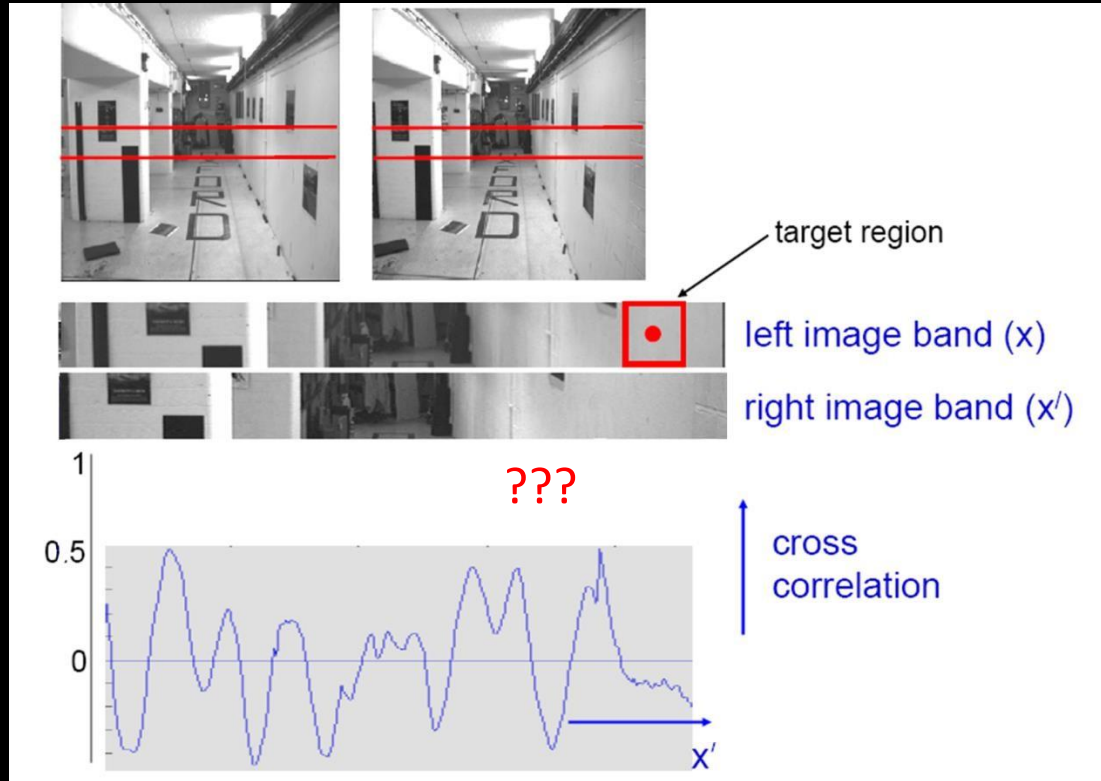


Correlation-based window matching



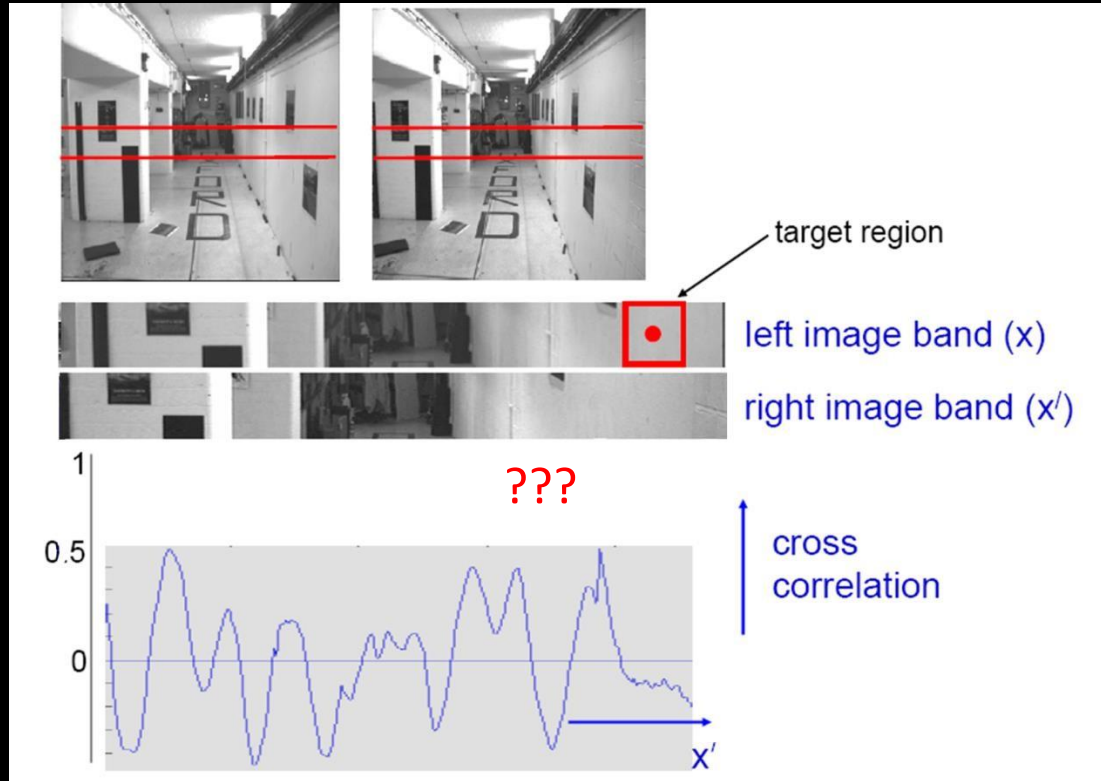
Texture less area

Correlation-based window matching



Where is the match?

Correlation-based window matching



Problem:
window is too
small to
capture
features

What are the good regions to Match?

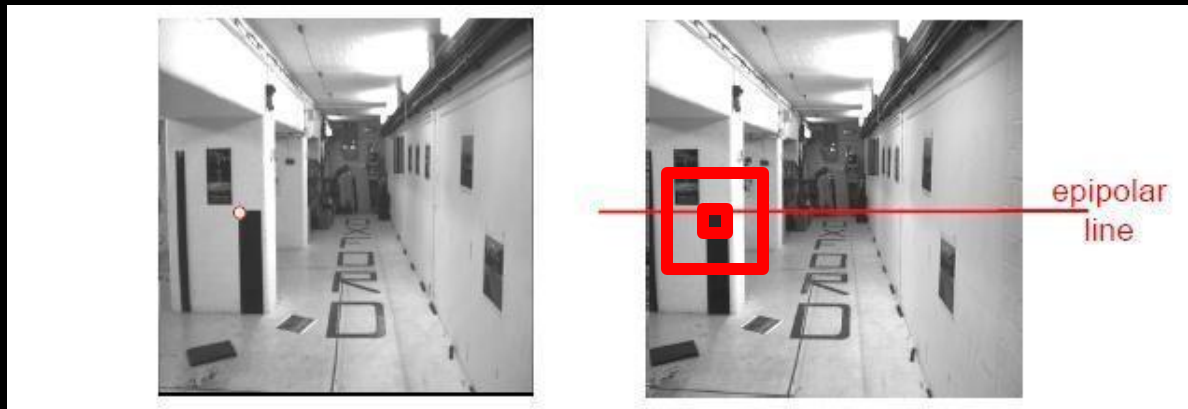


What are the good regions to Match?



C, D, E, K

Effect of window size



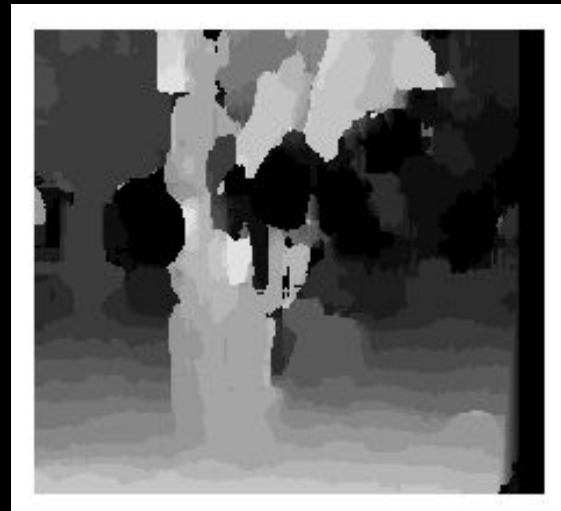
how do we pick the correct window size?

Source: Andrew Zisserman

Effect of window size



$W = 3$



$W = 20$

performing sliding
window

noise is there

Scale is always an
issue

Figures from Li Zhang

Correspondence problem

Beyond the hard constraint of epipolar geometry, there are “soft” constraints to help identify corresponding points

- Similarity
- Uniqueness
- Ordering
- Disparity gradient is limited

Uniqueness constraint

No more than one
match in right image
for every point in
left image

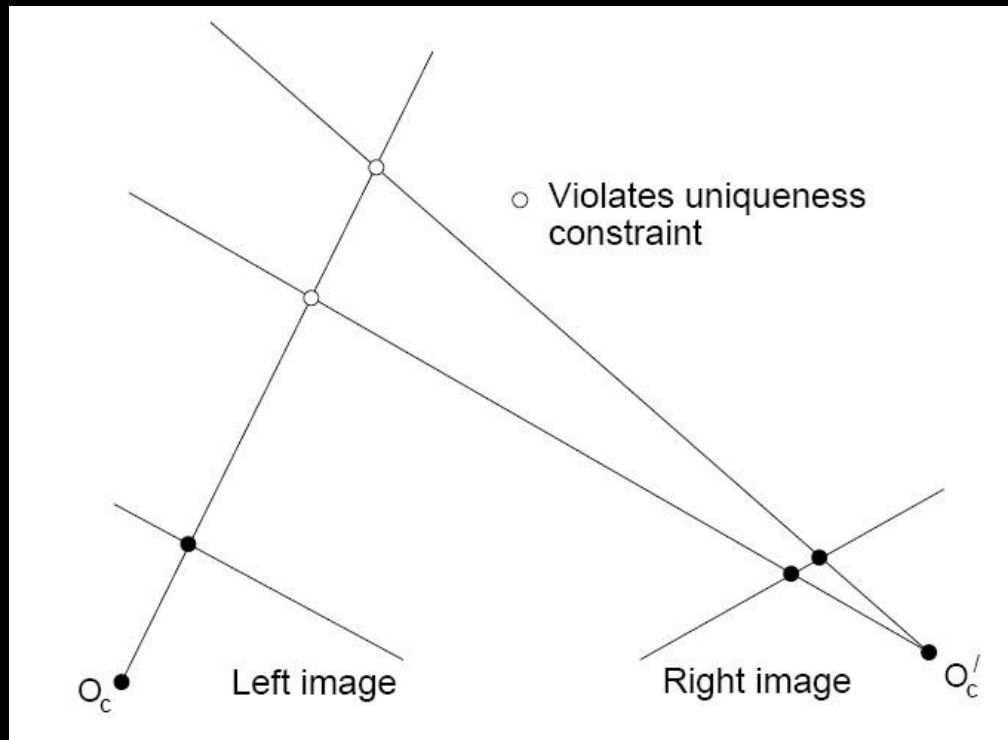
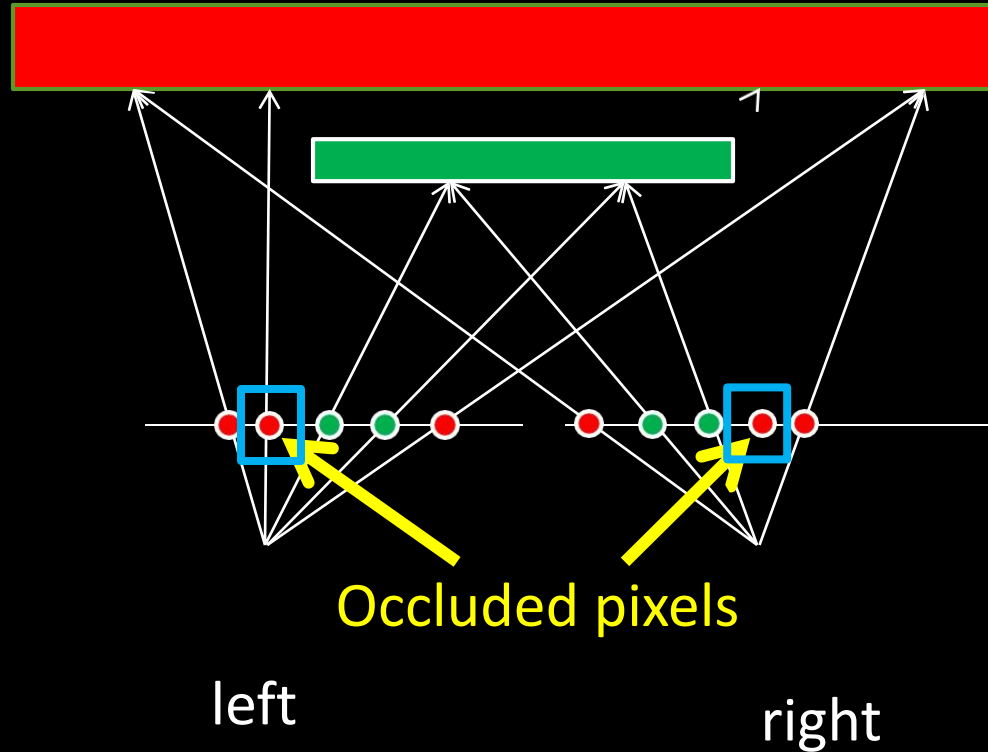


Figure from Gee & Cipolla 1999

Problem: Occlusion



Stereo results

Image data from University of
Tsuku'

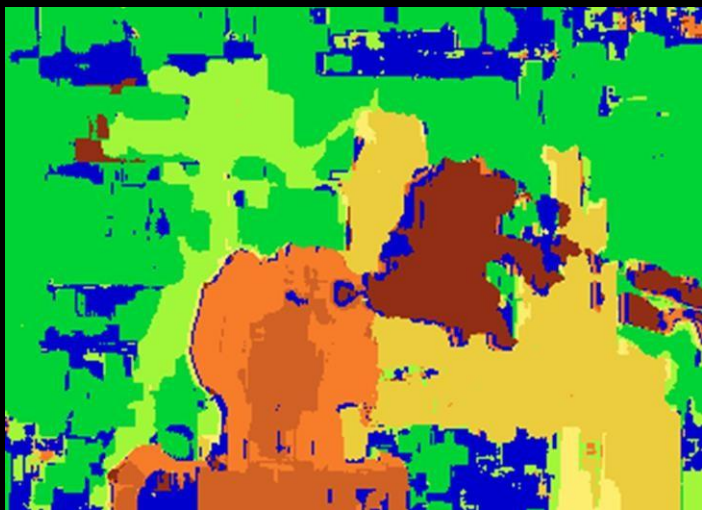


Scene



Ground truth : actual
depth

Results with window search



Window-based matching
(best window size)



Ground truth

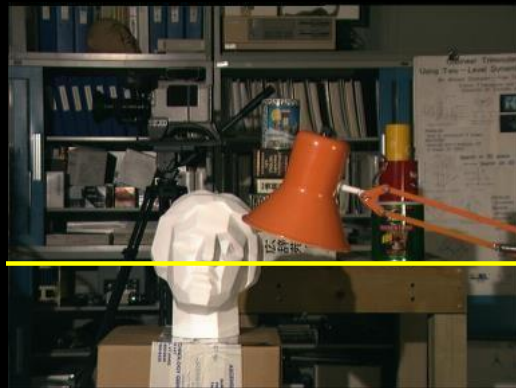
Better solutions

Beyond individual correspondences to estimate disparities:

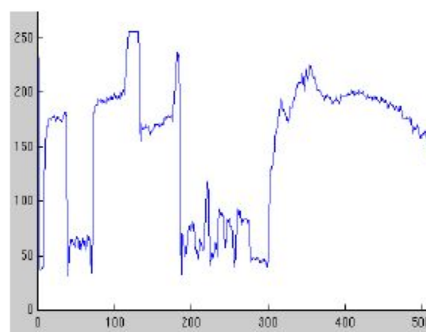
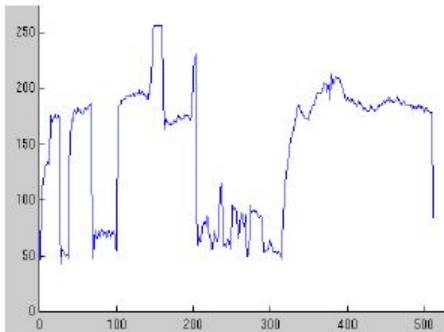
- Optimize correspondence assignments jointly
 - Scanline at a time (DP)
 - Full 2D grid (graph cuts)

Scanline stereo

Coherently match pixels on the entire scanline



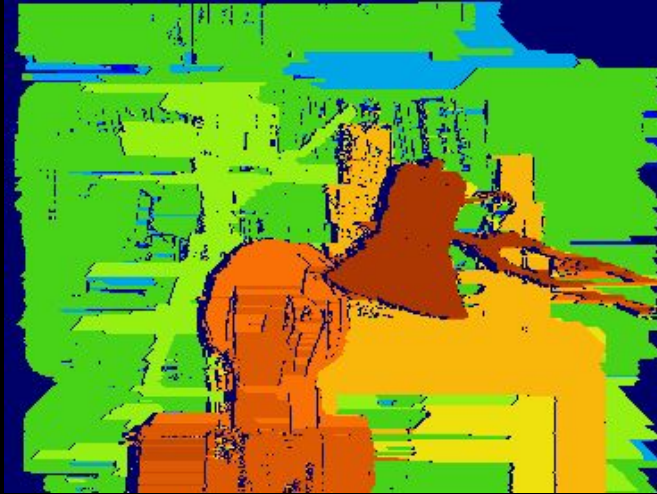
intensity



Goal is
to take
the best
match!

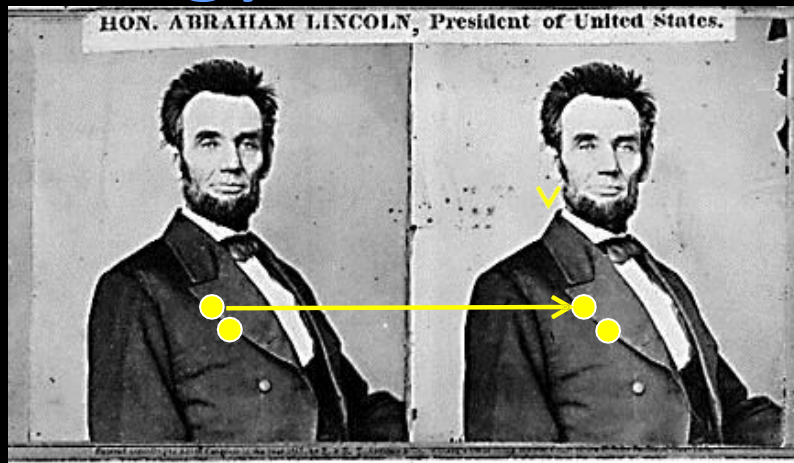
Coherent stereo on 2D grid

Scanline stereo generates streaking artifacts



Can't use dynamic programming to find spatially coherent disparities/ correspondences on a 2D grid

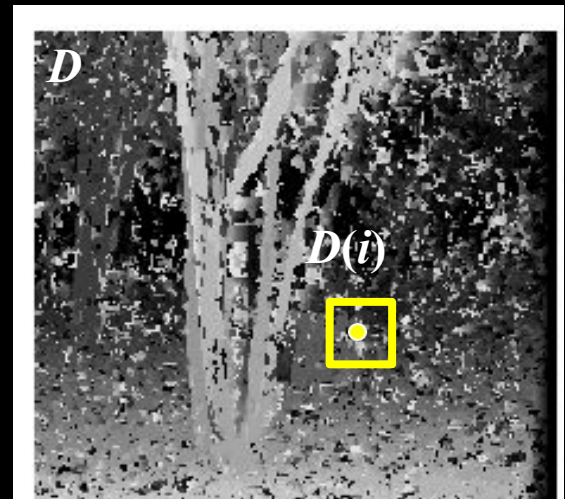
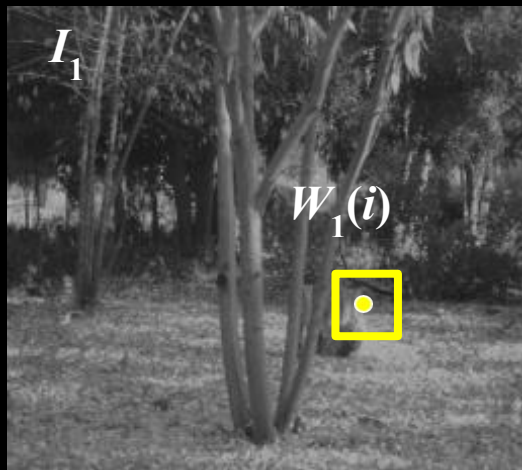
Stereo as energy minimization



What defines a good stereo correspondence?

1. **Match quality** - Want each pixel to find a good appearance match in the other image
2. **Smoothness** - if two pixels are adjacent, they should (usually) move about the same amount

Stereo matching as energy minimization

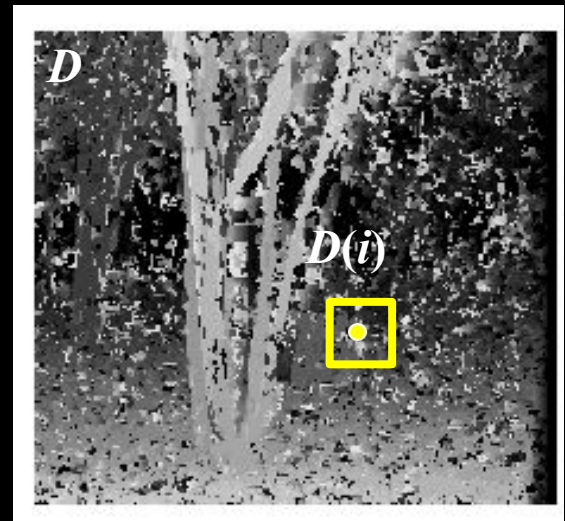
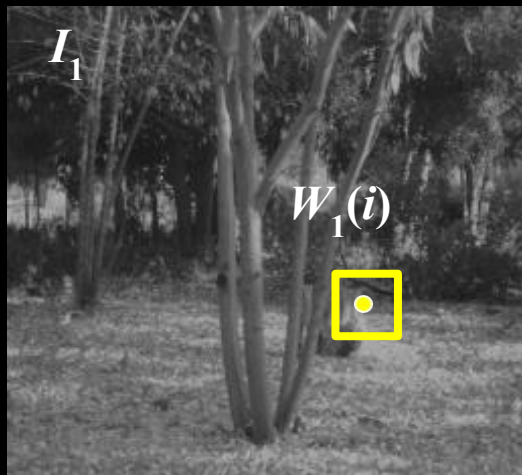


Data term:

$$E_{\text{data}} = \sum_i \left(W_1(i) - W_2(i + D(i)) \right)^2$$

Source: Steve Seitz

Stereo matching as energy minimization

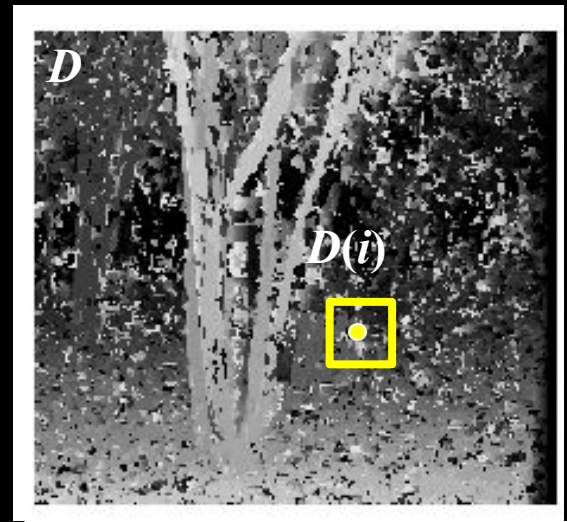
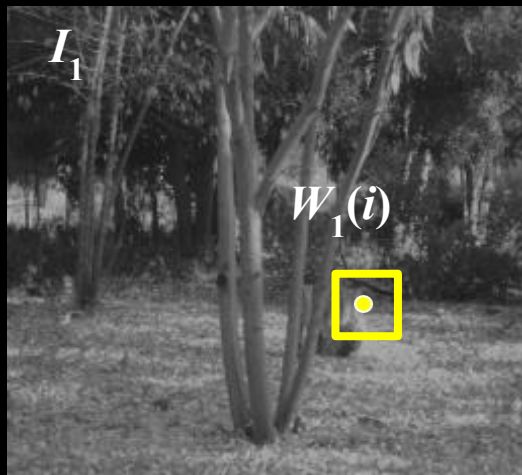


Smoothness
term:

$$E_{\text{smooth}} = \sum_{\text{neighbors } i, j} \rho(D(i) - D(j))$$

Source: Steve Seitz

Stereo matching as energy minimization



Total energy:

$$E = \alpha E_{\text{data}}(I_1, I_2, D) + \beta E_{\text{smooth}}(D)$$

Better results...

- Energy functions of this form can be minimized using *graph cuts*
- Y. Boykov, O. Veksler, and R. Zabih, **Fast Approximate Energy Minimization via Graph Cuts**, PAMI 2001

Better results...



State of the art method Ground truth

For the latest and greatest: <http://www.middlebury.edu/stereo>

Challenges

- Low-contrast ; textureless image regions
- Occlusions
- Violations of brightness constancy (e.g., specular reflections)
- Really large baselines (foreshortening and appearance change)
- Camera calibration errors